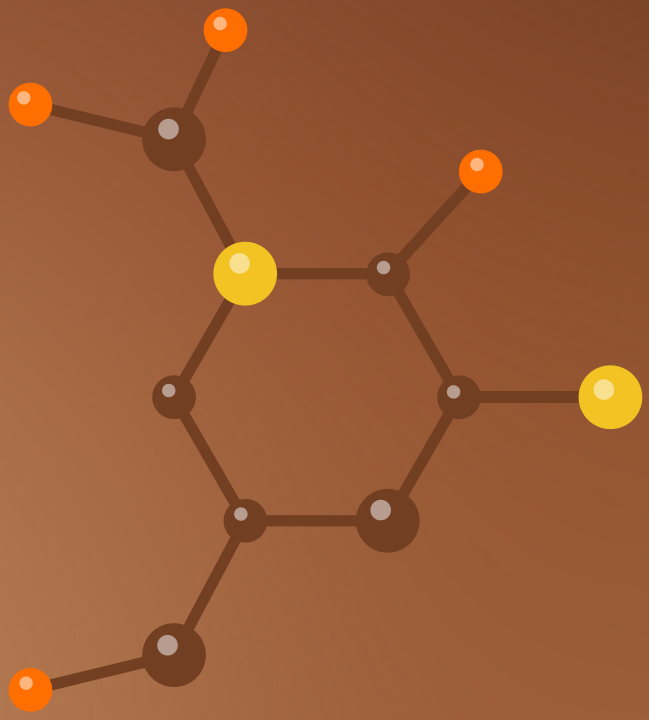
ALLEN CAREER INSTITUTE
path to success
KOTA (RAJASTHAN)



ORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Polymer & Practical Organic Chemistry, Separation Techniques

English Medium

EXERCISE # O-I

1. Which of the following is monomer unit of polystyrene :-

(A) $CH_2 = \underset{\underset{CN}{|}}{CH}$ (B) $CF_2=CF_2$ (C) $CH_2=\underset{\underset{Cl}{|}}{CH}$ (D) $\underset{\underset{C_6H_5}{|}}{CH}=CH_2$

PL0001

2. Weakest intermolecular forces are present in :-

(A) Neoprene (B) Terylene (C) Polystyrene (D) Bakelite

PL0002

3. Thermosetting polymer, Bakelite is formed by the reaction of phenol with :-

(A) CH_3CH_2CHO (B) CH_3CHO (C) HCHO (D) HCOOH

PL0003

4. Which one is classified as a condensation polymer?

(A) Teflon (B) Acrylonitrile (C) Dacron (D) Neoprene

PL0004

5. Novolac is a :

(A) linear polymer of urea and formaldehyde

(B) crosslink polymer of urea and formaldehyde

(C) linear polymer of phenol and formaldehyde

(D) crosslink polymer of phenol and formaldehyde

PL0005

6. Which of the following is not a semisynthetic polymer.

(A) cis-polyisoprene (B) cellulose nitrate

(C) cellulose acetate (D) vulcanised rubber

PL0006

7. Which one of the following polymers is prepared by condensation polymerization

(A) Styrene (B) Nylon–6,6 (C) Teflon (D) Rubber

PL0007

8. Which of the following statement is not true :

(A) Natural rubber has the trans-configuration at every double bond.

(B) Buna-S is a copolymer of butadiene and styrene.

(C) Natural rubber is a 1, 4-polymer of isoprene.

(D) In vulcanization, the formation of sulphur bridges between different chains make rubber harder and stronger.

PL0008

9. Structures of some common polymers are given. Which one is not correctly presented ?

(A) Nylon 66 $+NH(CH_2)_6NHCO(CH_2)_4 - CO -]_n$ (B) Teflon $+CF_2 - CF_2 -)_n$

(C) Neoprene $\left(- CH_2 - \underset{\underset{Cl}{|}}{C} = CH - CH_2 - CH_2 -\right)_n$ (D) Terylene $+OC-C_6H_4-COOCH_2-CH_2-O)_n$

PL0009

10. Of the following which one is classified as polyester polymer ?

(A) Terylene (B) Bakelite

(C) Malamine formaldehyde (D) Nylon-66

PL0010

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11. Which one of the following is not a condensation polymer?
(A) Dacron (B) Neoprene
(C) Melamineformaldehyde (D) Glyptal

PL0011

12. Which of the following statements is false?
(A) The repeat unit in natural rubber is isoprene
(B) Both starch and cellulose are polymers of glucose
(C) Artificial silk is derived from cellulose
(D) Nylon-66 is an example of elastomer

PL0012

13. Which one of the following sets forms the biodegradable polymer ?
(A) $HO - CH_2 - CH_2 - OH$ & HOOC-  -COOH
(B)  $-CH = CH_2$ and $CH_2 = CH - CH = CH_2$
(C) $CH_2 = CH - CN$ and $CH_2 = CH - CH = CH_2$
(D) $H_2N - CH_2 - COOH$ and $H_2N - (CH_2)_5$–COOH

PL0013

14. Nylon is an example of :-
(A) Polythene (B) Polyester (C) Polysaccharide (D) Polyamide

PL0014

15. Select the incorrect statement about Nylon 2-nylon-6.
(A) It is a copolymer
(B) It is biodegradable
(C) It is an alternating polyamide
(D) It is made up of $CH_3 - \underset{\displaystyle NH_2}{\underset{|}{CH}} - COOH$ and $H_2N(CH_2)_5COOH$

BM0036

16. The monomer that can undergo radical, cationic and anionic polymerisation with equal case-
(A) $Me-\underset{\displaystyle Me}{\underset{|}{C}}=CH_2$ (B) $Ph - CH = CH_2$ (C) $CH_2 = CH_2$ (D) $CH_2 = CH - CN$

BM0037

17. Which of the following statements are incorrect :
(A) Copolymer of 1,3-butandiene & acrylonitrile is Buna-S
(B) HDPE is obtained by Ziegler-Natta catalyst
(C) Polymer dacron can be polyester fabric
(D) Phenol & formaldehyde resin is called novolac

BM0039

18. The monomer that undergo radical polymerisation most easily is
(A) $CH_2=CH_2$ (B) $C_6H_5CH=CH_2$ (C) $CH_2=C\begin{matrix}Me\\Me\end{matrix}$ (D) $CH_3–CH=CH_2$

BM0042

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ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

19. Select the **correct** statement

(A) High density polythene is a linear polymer.

(B) Low density polythene is a branched chain polymer.

(C) Chain growth polymers are also known as addition polymer.

(D) Step growth polymer is also known condensation polymer. **BM0050**

20. Select the correct statement.

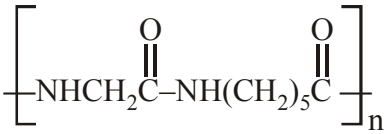
(A) Elastomers have the weakest intermolecular forces

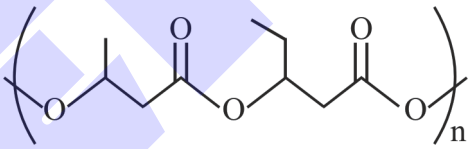
(B) Buna-N is example of synthetic copolymer

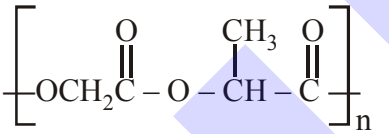
(C) Some fibres have crystalline nature

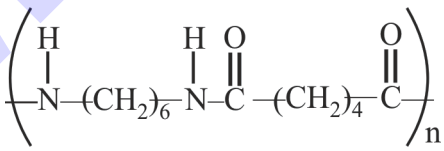
(D) Thermoplastic polymers have stronger intermolecular forces than fibres **BM0051**

21. Which of the following is not a biodegradable polymer?

(A) 

(B) 

(C) 

(D) 

PL0032

22. Which of the following is/are characteristic(s) of thermosetting polymers?

(A) Heavily branched cross linked polymers.

(B) Linear slightly branched long chain molecules.

(C) Become infusible on moulding so cann't be reused.

(D) Soften of heating and harden on cooling, can be reused. **PL0033**

23. Which of the following monomers form biodegradable polymers?

(A) 3-hydroxybutanoic acid + 3hydroxy pentanoic acid

(B) Glycine + Amino caproic acid

(C) Ethylene glycol + Phthalic acid

(D) Caprolactum **PL0034**

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24. Match the polymer in List-I with their main uses in List-II and choose the correct answer :

	List I (Name of polymer)		List II (Uses)
(A)	Polystyrene	(I)	Paints and lacquers
(B)	Glyptal	(II)	Rain coats
(C)	Polyvinyl chloride	(III)	Manufacture of toys
(D)	Bakelite	(IV)	Computer discs, handles of various untnsils.

The correct code is :

(A) A–II, B–I, C–III, D–IV (B) A–III, B–I, C–II, D–IV

(C) A–II, B–IV, C–III, D–I (D) A–III, B–IV, C–II, D–I **PL0035**

25. Match List I with List II :

	List I (Polymer)		List II (Used for items)
(A)	Nylon 6,6	(I)	Buckets, Dustbin
(B)	Low density polythene	(II)	Non-stick utensils
(C)	High density polythene	(III)	Bristles of brushes
(D)	Teflon	(IV)	Soft toys

Choose the correct answer from the options given below:

(A) A–III, B-I, C-IV, D-II (B) A–III, B-IV, C-I, D-II

(C) A–II, B-I, C-IV, D-III (D) A–II, B-IV, C-I, D-III **PL0036**

26. Match List I with List II :

	List I (Polymer)		List II (Used for)
(A)	Dextron	(I)	Protective clothing of fire-fighters.
(B)	Urea fromaldehyde	(II)	Stitching of wounds after surgical operation (Suture)
(C)	Nomex	(III)	Oil seals
(D)	Buna-N	(IV)	Wood laminates

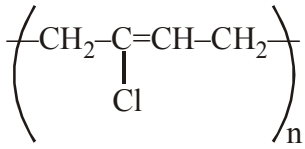
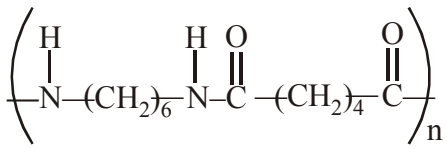
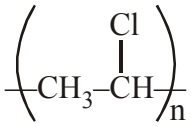
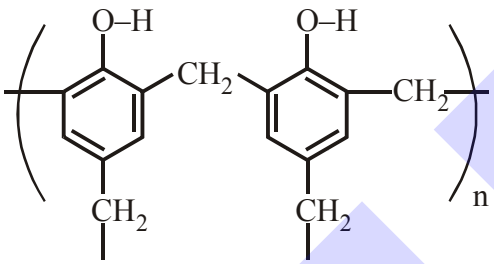
Choose the correct answer from the options given below:

(A) A–III, B-I, C-IV, D-II (B) A–III, B-IV, C-I, D-II

(C) A–II, B-I, C-IV, D-III (D) A–II, B-IV, C-I, D-III **PL0037**

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27. Match List-I with List-II

	List-I (Polymer)		List-II (Nature)
A.		I	Thermosetting polymer
B.		II	Fibers
C.		III	Elastomer
D.		IV	Thermoplastic polymer

Choose the correct answer from the options given below:

(A) A-II, B-III, C-IV, D-I
(B) A-III, B-II. C-IV, D-I
(C) A-III, B-I, C-IV, D-II
(D) A-I, B-III, C-IV, D-II

PL0038

28. Which of the following statements about low density polythene is false?

(A) It is a poor conductor of electricity.
(B) It's synthesis required dioxygen or a peroxide initiator as a catalyst.
(C) It is used in the manufacture of buckets, dust-bins etc.
(D) It's synthesis requires 1000 to 2000 atm pressure and 350 to 570 K temperature.

PL0039

29. The bakelite is made from phenol and formaldehyde. The initial reaction between the two compounds is an example of:

(A) Aromatic electrophilic substitution.
(B) Aromatic nucleophilic substitution.
(C) Free radical substitution reaction.
(D) Electrophilic addition reaction.

PL0040

30. Which of the following statement(s) is/are correct about Vulcanisation of rubber?

(A) This process consists of heating a mixture of raw rubber with sulphur at a temperature rang between 373 K to 415 K.
(B) On vulcanisation, sulphur forms cross links and thus the rubber gets stiffened.
(C) In the manufacture of tyre rubber 5% of sulphur is used as a crosslinking agent.
(D) All are incorrect

PL0041

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31. Which of the following statement(s) is/are true about thermoplastics?

(A) The intermolecular forces in thermoplastic polymers are weaker than the forces present in elastomers.

(B) There is no cross links between the polymeric chains.

(C) These are capable of repeatedly softening on heating and hardening on cooling.

(D) Polystyrene is a thermoplastic polymer. **PL0042**

32. Melamc, a thermosetting plastic often used to make non breakable plastic dishes, is prepared by the copolymerisation of :

(A) Urea and formaldehyde (B) Melamine and fomraldehyde

(C) Phenol and formaldehyde (D) Terephthalic acid and ethylene glycol

PL0043

33. In which case both polymers in pair have weakest intermolecular forces?

(A) Nylon-6, Dacron (B) PVC, Buna-S

(C) Natural rubber, Buna-N (D) Polystyrene, Polyvinyl cyanide **PL0044**

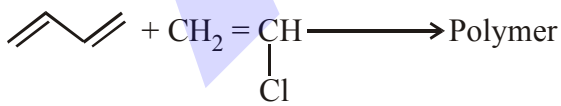
34. Match List I with List II

	List I (Name of polymer)		List II (Uses)
A.	Glyptal	I.	Flexible pipes
B.	Neoprene	II.	Synthetic wool
C.	Acrilan	III.	Paints and Lacquers
D.	LDP	IV.	Gaskets

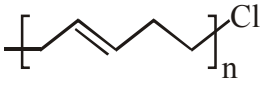
Choose the correct answer from the options given below :-

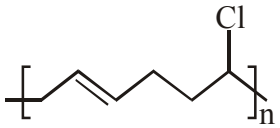
(1) A-III, B-II, C-IV, D-I (2) A-III, B-IV, C-II, D-I

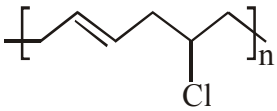
(3) A-III, B-IV, C-I, D-II (4) A-III, B-I, C-IV, D-II **PL0045**

35. 

The polymer obtained is:

(1) 

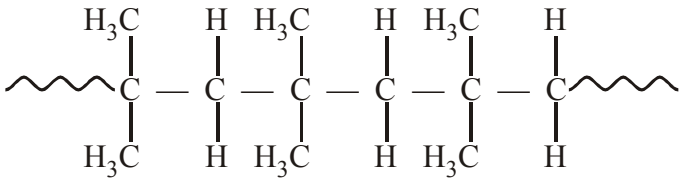
(2) 

(3) 

(4) All of these **PL0046**

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36. For the polymer



The structural formula for the monomer is :

(A) $CH_3–CH=CH_2$ (B) $CH_3–CH=CH–CH_3$

(C) $(CH_3)_2C = CH_2$ (D) $CH_3C \equiv CCH_3$ **PL0047**

37. Which of the following polymers chains are held together by hydrogen bonds?

(A) Neoprene (B) PMMA (C) Nylon–6, 6 (D) Teflon

PL0048

38. Which polymer is used to manufacture rain coats, hand bags, vinyl flooring, water pipes?

(A) Polystyrene (B) PVC (C) Propylene (D) Teflon

PL0049

39. Which polymer is used to make combs, electrical switches, handles of utensils and computer discs?

(A) Urea fromaldehyde resin (B) Bakelite

(C) Propylene (D) Nylon-6 **PL0050**

40. Which of the following is/are correctly matched?

(A) Neoprene → Chloroprene

(B) Natural rubber → Isoprene

(C) Teflon → Tetrafluoroethene

(D) Acrilan → Acrylonitirle **PL0051**

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EXERCISE # O-II (SEPRATION TECHNIQUES)

1. For the Separation of two immiscible liquids which method (or apparatus) is used?

(1) Chromatography (2) Fractionating column

(3) Fractional distillation (4) Separating funnel **PO0001**

2. A mixture of benzene and chloroform is separated by

(1) Sublimation (2) Separating funnel

(3) Crystallization (4) Distillation **PO0002**

3. Aniline is usually purified by

(1) Chromatographic technique (2) Steam distillation

(3) By addition of oxalic acid (4) Fractional crystallization **PO0004**

4. The most suitable method for separation of 1:1 mixture of ortho- and para- nitrophenols is

(1) Sublimation (2) Chromatography

(3) Crystallization (4) Steam distillation

PO0005

5. Distillation under reduced pressure is employed for

(1) Benzene (2) Petrol

(3) Glycerol (4) Organic compounds used in medicine **PO0006**

6. Impure glycerine is purified by

(1) Steam distillation (2) Simple distillation

(3) Vacuum distillation (4) None of the above **PO0007**

7. The latest technique used for purification for organic compuounds in

(1) Chromatography (2) Vacuum distillation

(3) Fractional distillation (4) Crystallisation **PO0009**

8. In paper chromatography

(1) Moving phase is liquid and stationary phase is solid

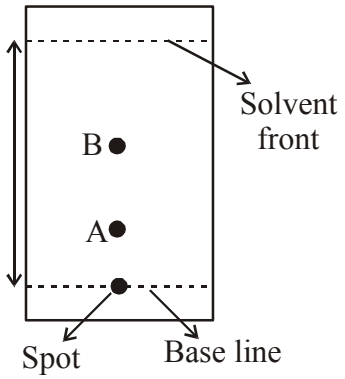
(2) Moving phase is liquid and stationary phase is liquid

(3) Moving phase is solid and stationary phase is solid

(4) Moving phase is solid and stationary phase is liquid **PO0010**

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9. Using the provided information in the following paper chromatogram :



Paper chromatography
for compounds A and B

Statement 1 : B is less polar than A.

Statement 2 : Non-polar solvent interact more with less polar substance and interact less with more polar substance.

(1) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(2) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(3) Statement-1 is true, statement-2 is false.

(4) Statement-1 is false, statement-2 is true. **PO0084**

10. The best method for the separation of naphthalene and benzoic acid from their mixture is

(1) Chromatography (2) Crystallisation (3) Distillation (4) Sublimation **PO0085**

11. A is a lighter phenol and B is an aromatic carboxylic acid. Separation of mixtue A and B can be carried out easily by using a solution of

(1) Sodium hydroxide (2) Sodium sulphate (3) Calcium chloride (4) Sodium bicarbonate **PO0011**

12. In sodium fusion test of organic compounds the nitrogen of the organic compound is converted into

(1) Sodamide (2) Sodium cyanide (3) Sodium nitrite (4) Sodium nitrate **PO0012**

13. The Lassaigne's extract is boiled with dil. HNO_3 before testing for halogens because

(1) Silver halides are soluble in HNO_3 (2) Na_2S and NaCN are decomposed by HNO_3

(3) Ag_2S is soluble in HNO_3 (4) AgCN is soluble in HNO_3 **PO0013**

14. The compound formed upon subjecting an aliphatic amine to Lassaigne's test is:

(1) $NaNH_2$ (2) $NaNO_2$ (3) NaCN (4) NaN_3 **PO0086**

15. In a Lessaigne's test for nitrogen, the blue colour is due to the formation of

(1) Potassium ferricyanide (2) Sodium cyanide

(3) Sodium ferrocyanide (4) Ferri-ferro cyanide **PO0014**

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16. Which of the following compounds does not show Lassaigne test for nitrogen?

(1) Urea (2) Hydrazine (3) Phenyl hydrozene (4) Azo benzene

PO0015

17. Acidified sodium fusion extract on addition of ferric chloride solution gives red colouration which confirms the presence of :

(1) Sulphur and chlorine (2) Nitrogen and sulphur

(3) Only nitrogen (4) Only sulphur **PO0087**

18. When piece of human hair are heated srongly with soda lime smell of ammonia can be detected. which one of the following conclusion can be drawn from this observation.

(1) Ammonia is present in human hair (2) Ammonium salt is present in hair

(3) Hair contains amino acids (4) None of the above **PO0016**

19. The Beilstein test for organic compounds is used to detect

(1) Nitrogen (2) Sulphur (3) Carbon (4) Halogen **PO0017**

20. In Kjeldahl's method, $CuSO_4$ acts as :

(1) Oxidising agent (2) Reducing agent

(3) Hydroylsing agent (4) Catalytic agent **PO0088**

21. Which of the following is the best scientific method to test presence of water in liquid?

(1) Smell (2) Taste

(3) Use of litmus paper (4) Use of anhydrous copper sulphate **PO0018**

22. Liebig's method is used for the estimation of

(1) Nitrogen (2) Sulphur

(3) Carbon and hydrogen (4) Halogens **PO0019**

23. 0.765 gm of an acid gives 0.535 gm of CO_2 and 0.138 gm of H_2O. Then the ratio of percentage of carbon and hydrogen is

(1) 19:2 (2) 18:11 (3) 20:17 (4) 1:7 **PO0020**

24. In a planar chromatographic analysis of organic compounds, [W, X, Y and Z] the solvent front was18.0 cm and the fonts due to the compounds W, X, Y and Z were 16.6, 14.3, 10.2 and 5.5 cm, respectively. If the R, value of the unknown compound is 0.79, identify the unknown compound among the W, X, Y and compounds.

(1) W (2) X (3) Y (4) Z **PO0089**

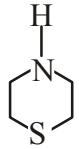
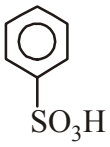
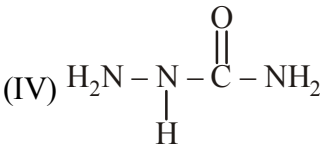
25. Nitrogen is an organic compound can be estimated by

(1) Kjeldahl's method only (2) Duma's method only

(3) Both the methods (4) none of these methods **PO0021**

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26. Which of the following compound(s) will give blue colour when it is converted into Lassaigne's extract and $FeSO_4$ is added followed by $FeCl_3$.

(I)  (II)  (III) NH_2–OH (IV) 

(1) I and IV (2) IV only (3) I, III and II (4) I, II, III and IV

PO0090

27. Dumas method involves the determination of nitrogen content in the organic compound in form of

(1) NH_3 (2) N_2 (3) NaCN (4) $(NH_4)_2SO_4$ **PO0022**

28. In Kjeldahl's method, nitrogen present is estimated as

(1) N_2 (2) NO (3) NH_3 (4) NO_2 **PO0023**

29. The percentage of sulphur in an organic compound whose 0.32g produces 0.233g of $BaSO_4$ [At.wt.Ba=137, S=32] is

(1) 1.0 (2) 10.0 (3) 23.5 (4) 32.1 **PO0024**

30. If 0.2 gram of an organic compound containing carbon, hydrogen and oxygen on combustion, yielded 0.147 gram carbon dioxide and 0.12 gram water. What will be the content of oxygen in the substance?

(1) 73.29% (2) 78.45% (3) 83.23% (4) 89.50% **PO0025**

31. Simple distillation can be used to separate liquids which differ in their boiling points at least by

(1) 5°C (2) 10°C (3) 35 - 50°C (4) 100°C **PO0036**

32. Molecular mass of a volatile substance may be obtained by

(1) Kjeldahl's method (2) Duma's method

(3) Victor meyer's method (4) Liebig's method **PO0047**

33. Kjeldahl's method can not be used for the estimation of Nitrogen in

(1) Pyridine (2) Nitro compounds

(3) Azo compounds (4) All **PO0049**

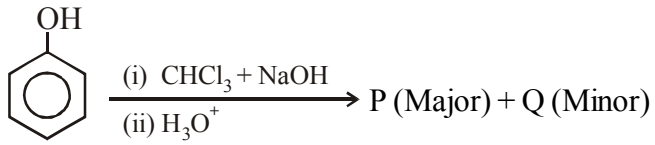
34. The ammonia evolved from the treatment of 0.30 g of an organic compound for the estimation of nitrogen was passed in 100 mL of 0.1 M sulphuric acid. The excess of acid required 20 mL of 0.5 M sodium hydroxide solution for complete neutralization. The organic compound is?

(1) Urea (2) Benzamide (3) Acetamide (4) Thiourea

PO0091

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35. Consider the reaction-



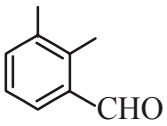
Mixture of A and B can be best separated by -

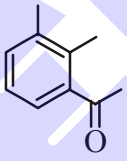
(1) Steam distillation (2) Vacuum distillation

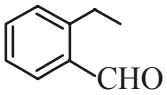
(3) Fractional distillation (4) Crystallisation

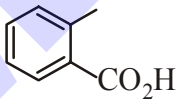
BM0038

36. An organic compound on reaction with 2,4-dinitrophenylhydrazine (2,4–DNP) gives a yellow precipitate. It also gives silver mirror on reaction with ammonical $AgNO_3$. It gives an alcohol and sodium salt of a carboxylic acid on reaction with concentrated NaOH. It yields benzene-1,2-dicarboxylic acid on heating with alkaline $KMnO_4$. The structure of the compound among the following is

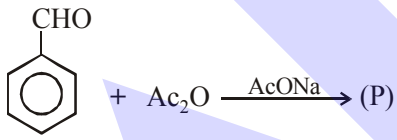
(1) 

(2) 

(3) 

(4) 

PO0092

37. 

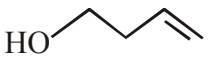
Before isolating (P) unreacted Ph–CHO is removed first. Select the correct statement.

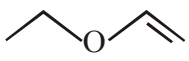
(1) P is cinnamaldehyde (2) P is crotonic acid

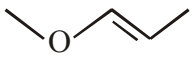
(3) Removal is done by simple distillation (4) P is cinnamic acid.

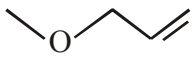
BM0053

38. An organic compound $P(C_4H_8O)$ is positive to Bayer's test but inert to sodium metal on treatment with conc. HCl, P gives CH_3CH_2Cl and CH_3CHO. The structure of P is?

(1) 

(2) 

(3) 

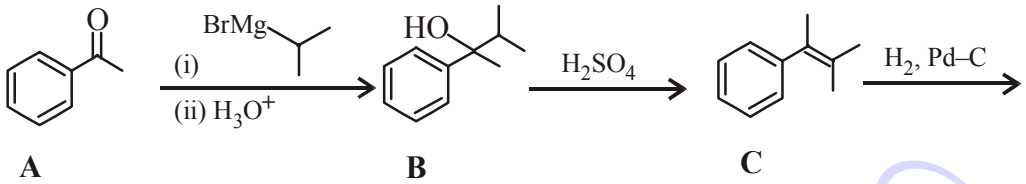
(4) 

PO0093

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39. A chemist carries out the synthetic scheme shown below. Unfortunately, the first two reactions are incomplete, and a mixture of compounds A, B and C is obtained after the second step. The chemist purfies this mixture by normal phase chromatography, using sillica gel as a stationary phase and a 10 : 1 hexane diethyl ether (v : v) solution as an eluent. In what order would compounds A, B and C elute off the column?

For eqch choice, the first compound to elute is listed first.

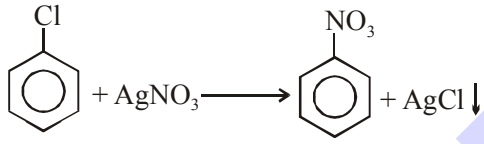


A **B** **C**

(1) C,A,B (2) A, C, B (3) B, A, C (4) C, B, A

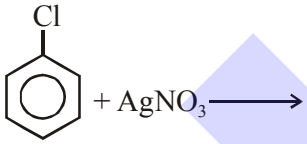
PO0094

40. Which of the following options are correct :

(1) 

(2) Na extract of sample containing chlorobenzene $\xrightarrow{+ AgNO_3}$ white ppt. of AgCl

(3) Na extract of sample containing chlorobenzene $\xrightarrow{+ AgNO_3}$ No ppt. of AgCl

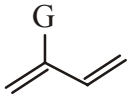
(4) 

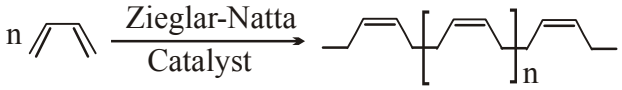
BM0052

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EXERCISE # S-I

Comprehension (Q.1 to Q.3)

The monomer  (G = Me or Cl) when treated with Zieglar - Natta catalyst undergo polymerisation in the manner given below -



cis-poly-1,3-butadiene

1. The Zieglar-Natta catalyst is

(A) $TiCl_4$ (B) R_3Al (C) $R_3Al / TiCl_4$ (D) $R_3B/TiCl_4$

BM0066

2. The polymer obtained when monomeric unit used is $CH_2=\underset{\displaystyle Cl}{\underset{|}{C}}-CH=CH_2$

(A) Neoprene (B) Stilbene (C) Styrene (D) Chloropicrin

BM0067

3. Which of the following statement is not true considering the process given above.

(A) The general class of polymer formed is known as homopolymer

(B) The polymer obtained is stereoregular

(C) Buna–N can be prepared using above process

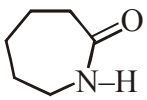
(D) Synthetic rubber can be formed by above process using 1,3- butadiene. **BM0068**

4. Match the column :

Column I	**Column II**
(A) Addition polymer	(P) Buna-S
(B) Condensation polymer	(Q) Buna-N
(C) Homopolymer	(R) Polythene
(D) Copolymers	(S) Nylon 6,6

BM0073

5. (a) What is the structure of nylon-6, made by alkaline polymerisation of caprolactum.



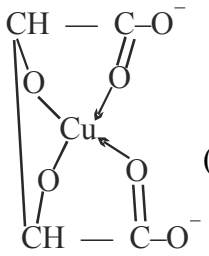
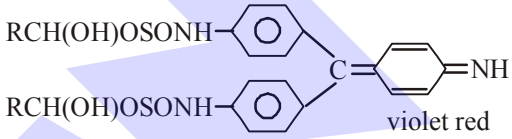
BM0103

(b) Suggest mechanism for the process of polymerisation of the chain growth and step growth reaction.

BM0103

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6. Match the column

Column I (Functional group)	**Column II (Test used or complex formed during confirmatory test)**
(A) Aldehydic	(P) $[(C_6H_5O)_6Fe]^{-3}$ (violet)
(B) Phenolic	(Q)  (blue)
(C) Alcohol	(R)  violet red
(D) α-D- Glucopyranose	(S) $(ROH)_2Ce(NO_3)_4$ (Red)
	(T) Molisch's Test

BM0072

7. Match the column

Column I (Component of mixture)	**Column II (Reagent)**
(A) Crystalline Na_2CO_3 + Sodium citrate + $CuSO_4$(aq. sol.)	(P) Fehling solution
(B) $CuSO_4$ + Rochelle Salt + NaOH(Aq. sol.)	(Q) Nesseler's Reagent
(C) 10% α - naphthol in alcohol	(R) Bennedict's solution
(D) HgCl + KI + KOH (aq. sol.)	(S) Molisch's Reagent

BM0059

8. Compound (A) $C_5H_{10}O_5$ give a tetra-acetate with Ac_2O & oxidation of A with Br_2/H_2O gives an acid $C_5H_{10}O_6$. Reduction of A with HI & Red P. Give 2-methyl butane. What is structure of 'A'

BM0104

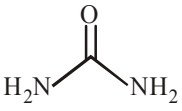
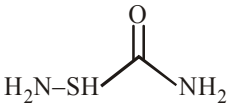
9. In Duma's method for estimation of nitrogen, 0.3 g of an organic compound gas 50 mL of N_2 collected at 300 K temperature and 715 mmHg pressure. Calculate the % compostition of nitrogen in compound. (Aq. Tension 15 mmHg) **PO0095**

10. A sample of 0.40 g of an organic compound was treated according to Kjeldahl's method. The NH_3 evolved was absorbed in 50 mL of 0.5 MH_2SO_4. The residual acid required 60 mL of 0.5 M solution of NaOH for neutralization. Find the % of N in compound. **PO0096**

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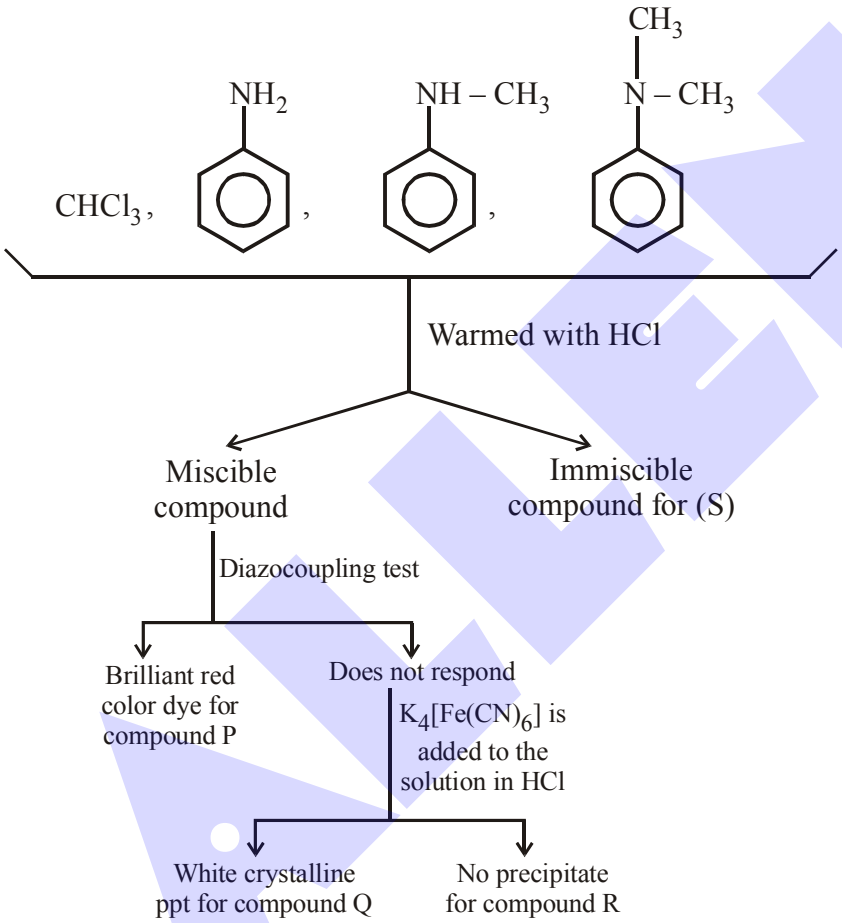
11. Which of the following compound gives blood red ppt when their sodium extract solution is treated with neutral $FeCl_3$ Solution?

(i) Cysteine (ii) Methionine (iii) Cystine (iv) Glycine (v) 

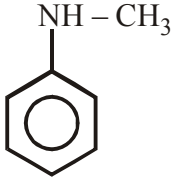
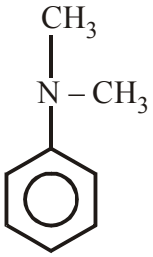
(vi)  (vii)  (viii)  (ix) Penicilline

PO0097

Paragraph based question 12 to 14



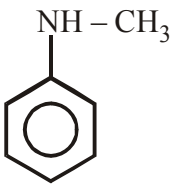
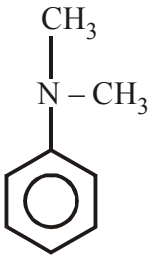
12. Compound P is

(A) $CHCl_3$ (B)  (C)  (D) 

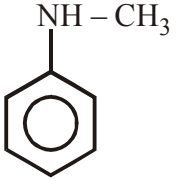
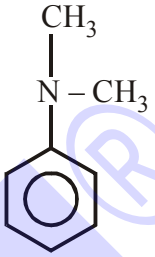
PO0098

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13. Compound Q is

(A) $CHCl_3$ (B) Aniline (C)  (D)  **PO0099**

14. Compound S is :

(A) $CHCl_3$ (B)  (C)  (D)  **PO0100**

15. Which of the following statement(s) is/are correct about Kevlar polymer?
(A) Kevlar is an aramid.
(B) It is used for making army helmets and lightweight bullet proof vests
(C) It is prepared by condensation polymerisation
(D) It is a copolymer of terephthalic acid and p-phenylene diamine.
PL0052

16. Which of the following statement(s) is/are correct about Lexan polymer?
(A) It is a polycarbonate polymer.
(B) Lexan is a strong and transparent polymer.
(C) It is used for bulletproof windows and traffic light lenses.
(D) It is prepared by the reaction of phosgene with bisphenol A. **PL0053**

17.

Column-I	**Column-II**
(A) Natural rubber	(P) Poly β-glucose, addition polymer
(B) Cellulose	(Q) Co-polymer, condensation polymer
(C) Nylone-6	(R) Poly cis-2-methyl-1,3-butadiene
(D) Dacron	(S) Poly β-D (+) glucose, condensation polymer
	(T) Caprolactum, homopolymer

PL0054

18. How many of the following are prepared by condensation reaction?

(i) Dacron	(ii) Nylon-6,6	(iii) Nylon-6	(iv) Bakelite
(v) PMMA	(vi) Neoprene	(vii) Buna-N	(viii) PVC
(ix) Teflon			

PL0055

19. Count number of polymers which are step growth polymers as well as co-polymers :

(1) Polythene	(2) Nylon-6, 6	(3) Nylon-6, 10	(4) Dacron
(5) Bakelite	(6) Melamine formaldehyde resin		(7) Polyisoprene
(8) Styrene-butadiene rubber (Buna-S)		(9) PAN	(10) Teflon

PL0056

20. A segment of Nylon-6 have molecular mass 809 unit. The number of monomeric units present in the segment is : **PL0057**

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EXERCISE # J-MAIN

1. The distillation technique most suited for separating glycerol from spent-lye in the soap industry is **[J-Main-2016]**

(1) Distillation under reduced pressure (2) Simple distillation

(3) Fractional distillation (4) Steam distillation

BM0119

2. The formation of which of the following polymers involves hydrolysis reaction ? **[J-Main-2017]**

(1) Nylon 6 (2) Bakelite (3) Nylon 6, 6 (4) Terylene

BM0122

3. Which of the following statement is not true :- **[J-Main-2018]**

(1) Nylon 6 is an example of step-growth polymerisation

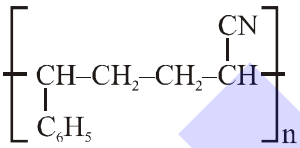
(2) Chain growth polymerisation involves homopolymerisation only

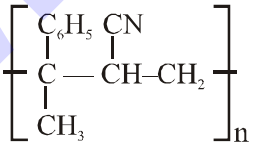
(3) Step growth polymerisation requires a bifunctional monomer

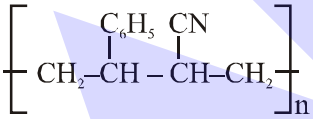
(4) Chain growth polymerisation includes both homopolymerisation and copolymerisation

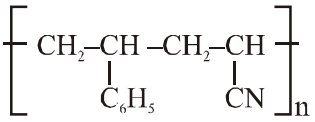
BM0124

4. The copolymer formed by addition polymerization of styrene and acrylonitrile in the presence of peroxide is : **[J-Main-2018]**

(1) 

(2) 

(3) 

(4) 

BM0126

5. Given below are two statements : **[JEE MAIN 2019]**

Statement I : A mixture of chloroform and aniline can be separated by simple distillation.

Statement II : When separating aniline from a mixture of aniline and water by steam distillation aniline boils below its boiling point.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement-I is false but statement II is true (2) Both statement-I and statement II are false

(3) Statement-I is true but statement II is false (4) Both statement-I and statement II are true

PO0053

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6. The correct match between item I and item II is : **[JEE MAIN 2019]**

	Item-I (mixture)		**Item-II (sepration method)**
(A)	H_2O : sugar	(P)	Sublimation
(B)	H_2O : Aniline	(Q)	Recrystallization
(C)	H_2O : Toluene	(R)	Steam disillation
		(S)	Differential extraction

(1) A → S, B → R, C → P

(2) A → Q, B → R, C → P

(3) A → R, B → P, C → S

(4) A → Q, B → R, C → S **PO0054**

7. If dichloromethane (DCM) and water (H_2O) are used for differential extraction, which one of the following statements is correct? **[JEE MAIN 2019]**

(1) DCM and H_2O would stay as upper and lower layer respectively in the separating funnel (S.F.)

(2) DCM and H_2O would stay as lower and upper layer respectively in the S.F.

(3) DCM and H_2O will make turbid / colloidal mixture.

(4) DCM and H_2O will be miscible clearly **PO0055**

8. The principle of column chromatography is : **[JEE MAIN 2019]**

(1) Gravitational force

(2) Capillary action

(3) Differential adsorption of the substances on the solid phase

(4) Differential absorption of the substances on the solid phase **PO0056**

9. In chromatography, which of the following statements is incorrect for R_f?

(1) R_f value depends on the type of chromatography. **[JEE MAIN 2019]**

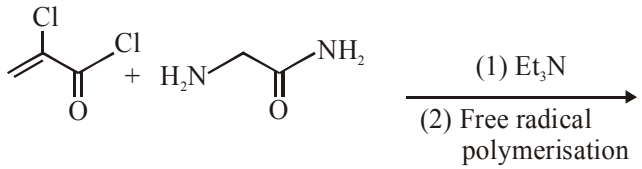
(2) Higher R_f value means higher adsorption.

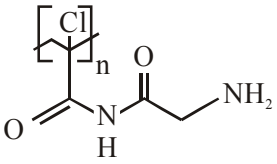
(3) R_f value is dependent on the mobile phase.

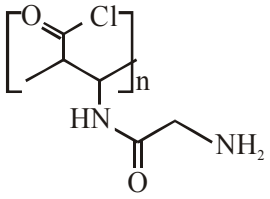
(4) The value of R_f can not be more than one. **PO0057**

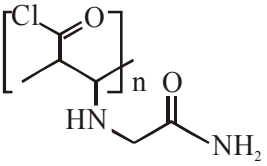
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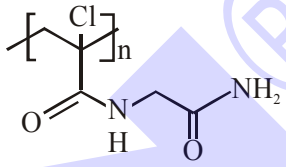
10. Major product of the following reaction is : **[J-Main-(Jan)-2019]**



(1) 

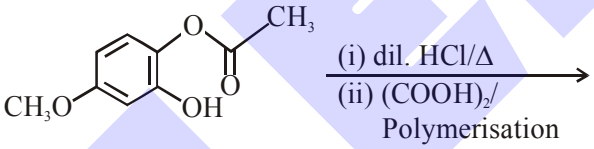
(2) 

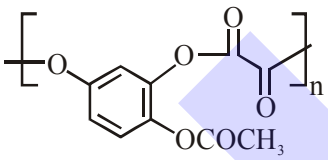
(3) 

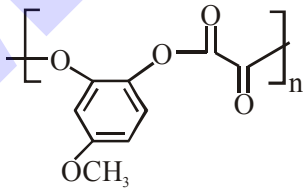
(4) 

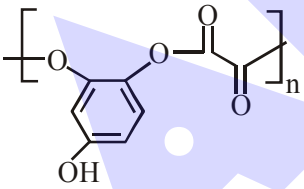
BM0131

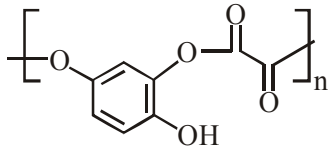
11. The major product of the following reaction is: **[J-Main-(Jan)-2019]**



(1) 

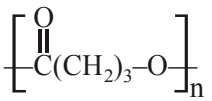
(2) 

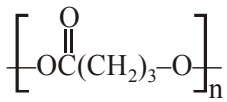
(3) 

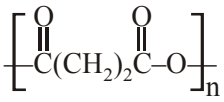
(4) 

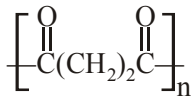
BM0132

12. The homopolymer formed from 4-hydroxy-butanoic acid is :- **[J-Main-(Jan)-2019]**

(1) 

(2) 

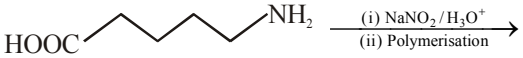
(3) 

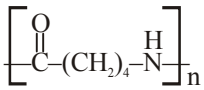
(4) 

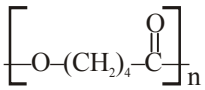
BM0134

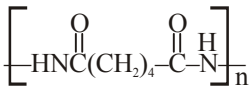
node06\B0BC-BD\Kota\JEE(Advanced)\Enfuse\Chem\Module\S-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\4-Poly & POC\03_Poly-POC-Sep_Exp65

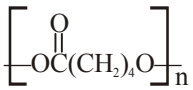
13. The polymer obtained from the following reactions is : **[J-Main-(Jan)-2019]**



(1) 

(2) 

(3) 

(4) 

BM0136

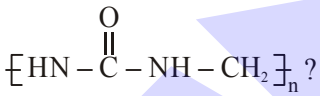
14. The two monomers for the synthesis of Nylone 6, 6 are : **[J-Main-(Jan)-2019]**

(1) $HOOC(CH_2)_6COOH$, $H_2N(CH_2)_6NH_2$ (2) $HOOC(CH_2)_4COOH$, $H_2N(CH_2)_4NH_2$

(3) $HOOC(CH_2)_6COOH$, $H_2N(CH_2)_4NH_2$ (4) $HOOC(CH_2)_4COOH$, $H_2N(CH_2)_6NH_2$

BM0138

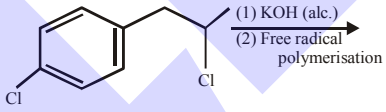
15. Which of the following compounds is a constituent of the polymer **[J-Main-(April)-2019]**

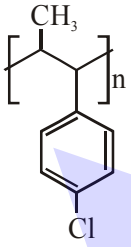


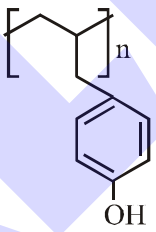
(1) Formaldehyde (2) Ammonia (3) Methylamine (4) N-Methyl urea

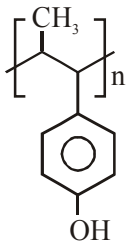
BM0146

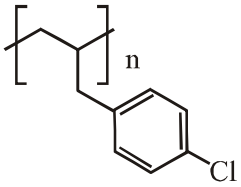
16. The major product of the following reaction is : **[J-Main-(April)-2019]**



(1) 

(2) 

(3) 

(4) 

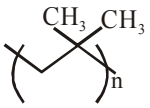
BM0147

17. Which of the following is a condensation polymer ? **[J-Main-(April)-2019]**

(1) Buna - S (2) Nylon 6, 6 (3) Teflon (4) Neoprene

BM0148

18. The correct name of the following polymer is: **[J-Main-(April)-2019]**



(1) Polyisoprene (2) Polytert-butylene (3) Polyisobutane (4) Polyisobutylene

BM0149

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19. Which of the following is a thermosetting polymer? **[J-Main-(April)-2019]**

(1) Buna–N (2) PVC (3) Bakelite (4) Nylon 6

BM0150

20. In chromotography technique, the purification of compound is independent of : **[JEE MAIN 2020]**

(1) Mobility or flow of solvent system (2) Solubility of the compound

(3) Length of the column or TLC Plate (4) Physical state of the pure compound

PO0059

21. Consider the Assertion and Reason given below. **[JEE MAIN 2020]**

Assertion (A) : Ethene polymerized in the presence of Ziegler Natta Catalyst at high temperature and pressure is used to make buckets and dustbins.

Reason (R): High density polymers are closely packed and are chemically inert. Choose the correct answer from the following :

(1) (A) is correct but (R) is wrong.

(2) (A) and (R) both are wrong.

(3) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A). **PL0015**

22. The correct match between **Item-I** and **Item-II** : **[JEE MAIN 2020]**

Item-I	**Item-II**
(a) Natural rubber	(I) 1, 3-butadiene + styrene
(b) Neoprene	(II) 1, 3-butadiene + acrylonitrile
(c) Buna-N	(III) Chloroprene
(d) Buna-S	(IV) Isoprene

(1) (a) - (III), (b) - (IV), (c) - (I), (d) - (II)

(2) (a) - (IV), (b) - (III), (c) - (II), (d) - (I)

(3) (a) - (IV), (b) - (III), (c) - (I), (d) - (II)

(4) (a) - (III), (b) - (IV), (c) - (II), (d) - (I)

PL0017

23. Match the following : **[JEE MAIN 2020]**

Test/Method	**Reagent**
(i) Lucas Test	(a) $C_6H_5SO_2Cl$/aq. KOH
(ii) Dumas method	(b) $HNO_3/AgNO_3$
(iii) Kjeldahl's method	(c) CuO/CO_2
(iv) Hinsberg Test	(d) Conc. HCl and $ZnCl_2$
	(e) H_2SO_4

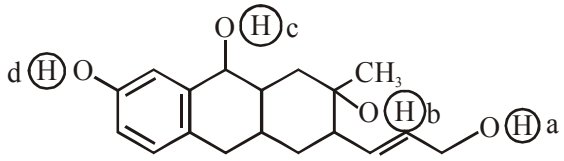
(1) (i)-(d), (ii)-(c), (iii)-(e), (iv)-(a) (2) (i)-(b), (ii)-(d), (iii)-(e), (iv)-(a)

(3) (i)-(d), (ii)-(c), (iii)-(b), (iv)-(e) (4) (i)-(b), (ii)-(a), (iii)-(c), (iv)-(d)

PO0070

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24. Consider the following reaction : **[JEE MAIN 2020]**



$\xrightarrow[\text{anhydride}]{\text{Chromic}}$ 'P'

The product 'P' gives positive ceric ammonium nitrate test. This is because of the presence of which of these –OH group(s) ?

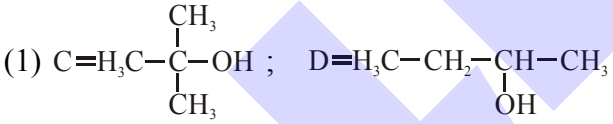
(1) (c) and (d) (2) (b) only (3) (d) only (4) (b) and (d)

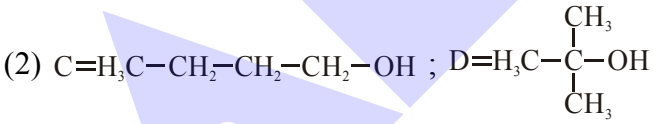
PO0071

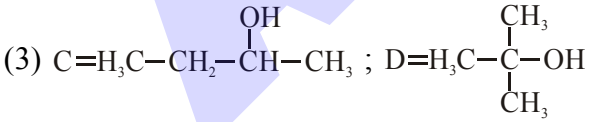
25. Two compounds A and B with same molecular formula (C_3H_6O) undergo Grignard's reaction with methylmagnesium bromide to give products C and D. products C and D show following chemical tests. **[JEE MAIN 2020]**

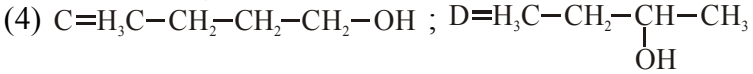
Test	C	D
Ceric ammonium nitrate Test	Positive	Positive
Lucas Test	Turbidity obtained after five minutes	Turbidity obtained immediately
Iodoform Test	Positive	Negative

C and D respectively are :

(1) 

(2) 

(3) 

(4) 

PO0072

26. Which one of the following set of elements can be detected using sodium fusion extract ?

(1) Sulfur, Nitrogen, Phosphorous, Halogens **[JEE MAIN 2021]**

(2) Phosphorous, Oxygen, Nitrogen, Halogens

(3) Nitrogen, Phosphorous, Carbon, Sulfur

(4) Halogens, Nitrogen, Oxygen, Sulfur **PO0063**

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27. In Carius method, halogen containing organic compound is heated with fuming nitric acid in the presence of : **[JEE MAIN 2021]**

(1) HNO_3 (2) $AgNO_3$ (3) $CuSO_4$ (4) $BaSO_4$ **PO0064**

28. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R) : **[JEE MAIN 2021]**

Assertion (A) : A simple distillation can be used to separate a mixture of propanol and propanone.

Reason (R) : Two liquids with a difference of more than 20°C in their boiling points can be separated by simple distillations.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) (A) is false but (R) is true.

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A) **PO0065**

29. Match List I with List II. **[JEE MAIN 2021]**

List I (Monomer Unit)	**List II (Polymer)**
(a) Caprolactum	(i) Natural rubber
(b) 2-Chloro-1,3-butadiene	(ii) Buna-N
(c) Isoperene	(iii) Nylon 6
(d) Acrylonitrile	(iv) Neoprene

Choose the correct answer from the options given below :

(1) (a) → (iv), (b) → (iii), (c) → (ii), (d) → (i)

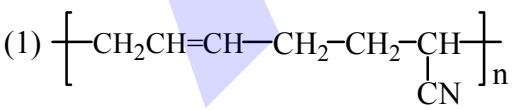
(2) (a) → (ii), (b) → (i), (c) → (iv), (d) → (iii)

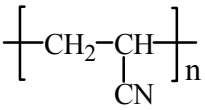
(3) (a) → (iii), (b) → (iv), (c) → (i), (d) → (ii)

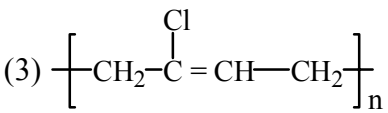
(4) (a) → (i), (b) → (ii), (c) → (iii), (d) → (iv)

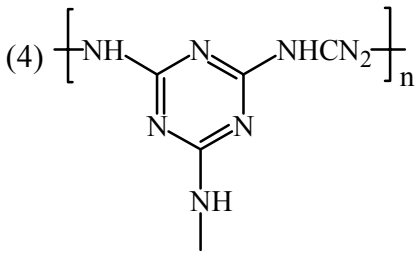
PL0018

30. The structure of Neoprene is - **[JEE MAIN 2021]**

(1) 

(2) 

(3) 

(4) 

PL0020

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31. Which of the following is 'a' FALSE statement? **[JEE MAIN 2021]**

(1) Carius tube is used in the estimation of sulphur in an organic compound

(2) Carius method is used for the estimation of nitrogen in an organic compound

(3) Phosphoric acid produced on oxidation of phosphorus present in an organic compound is precipitated as $Mg_2P_2O_7$ by adding magnesia mixture.

(4) Kjeldahl's method is used for the estimation of nitrogen in an organic compound

PO0073

32. Which of the following polymer is used in the manufacture of wood laminates ? **[JEE MAIN 2021]**

(1) *cis*-poly isoprene

(2) Melamine formaldehyde resin

(3) Urea formaldehyde resin

(4) Phenol and formaldehyde resin

PL0021

33. The functional groups that are responsible for the ion-exchange property of cation and anion exchange resins, respectively, are : **[JEE MAIN 2021]**

(1) $-SO_3H$ and $-NH_2$

(2) $-SO_3H$ and $-COOH$

(3) $-NH_2$ and $-COOH$

(4) $-NH_2$ and $-SO_3H$

PO0074

34. Match List-I with List-II : **[JEE MAIN 2021]**

List-I **(Chemicals)**	**List-II** **(Use / Preparation / Constituent)**
(a) Alcoholic potassium hydroxide	(i) Electrodes in batteries
(b) Pd/ $BaSO_4$	(ii) Obtained by addition reaction
(c) BHC (Benzene hexachloride)	(iii) Used for β- elimination reaction
(d) Polyacetylene	(iv) Lindlar's catalyst

Choose the most appropriate match :

(1) a-ii, b-i, c-iv, d-iii

(2) a-iii, b-iv, c-ii, d-i

(3) a-iii, b-i, c-iv, d-ii

(4) a-ii, b-iv, c-i, d-iii

PL0022

35. Bakelite is a cross-linked polymer of formaldehyde and : **[JEE MAIN 2021]**

(1) PHBV

(2) Buna-S

(3) Novolac

(4) Dacron

PL0023

36. Orlon fibres are made up of : **[JEE MAIN 2021]**

(1) Polyacrylonitrile

(2) Polyesters

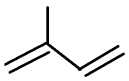
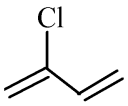
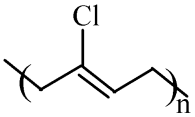
(3) Polyamide

(4) Cellulose

PL0024

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37. Match List-I with List-II : **[JEE MAIN 2021]**

List-I	List-II
(a) Chloroprene	(i) 
(b) Neoprene	(ii) 
(c) Acrylonitrile	(iii) 
(d) Isoprene	(iv) CH_2=CH-CN

Choose the correct answer from the options given below :

(1) (a) - (iii), (b)-(iv), (c) -(ii), (d) -(i) (2) (a) - (ii), (b)-(iii), (c) -(iv), (d) -(i)

(3) (a) - (ii), (b)-(i), (c) -(iv), (d) -(iii) (4) (a) - (iii), (b)-(i), (c) -(iv), (d) -(ii)

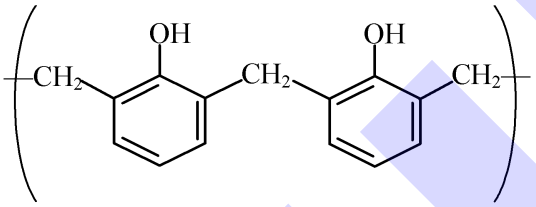
PL0025

38. A biodegradable polyamide can be made from: **[JEE MAIN 2021]**

(1) Glycine and isoprene (2) Hexamethylene diamine and adipic acid

(3) Glycine and aminocaproic acid (4) Styrene and caproic acid

PL0026

39.  **[JEE MAIN 2021]**

is a repeating unit for :

(1) Novolac (2) Buna-N (3) Acrilan (4) Neoprene

PL0027

40. Monomer units of Dacron polymer are : **[JEE MAIN 2021]**

(1) ethylene glycol and phthalic acid (2) ethylene glycol and terephthalic acid

(3) glycerol and terephthalic acid (4) glycerol and phthalic acid

PL0028

41. Ceric ammonium nitrate and $CHCl_3$ / alc. KOH are used for the identification of functional groups present in _____ and ___ respectively. **[JEE MAIN 2021]**

(1) Alcohol, phenol (2) Amine, alcohol (3) Alcohol, amine (4) Amine, phenol

PO0077

42. In chromotography technique, the purification of compound is independent of : **[JEE MAIN 2021]**

(1) Mobility or flow of solvent system (2) Solubility of the compound

(3) Length of the column or TLC Plate (4) Physical state of the pure compound

PO0078

43. Primary, secondary and tertiary amines can be separated using :- **[JEE MAIN 2021]**

(1) Para-Toluene sulphonyl chloride (2) Chloroform and KOH

(3) Benzene sulphonic acid (4) Acetyl amide

PO0079

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44. An organic compound "A" on treatment with benzene sulphonyl chloride gives compound B. B is soluble in dil. NaOH solution. Compound A is : **[JEE MAIN 2021]**

(1) C_6H_5–N–$(CH_3)_2$

(2) C_6H_5–NHCH$_2$CH$_3$

(3) C_6H_5–CH$_2$ NHCH$_3$

(4) C_6H_5–CH(CH$_3$)–NH$_2$

PO0081

45. Metallic sodium does not react normally with : **[JEE MAIN 2021]**

(1) gaseous ammonia (2) But-2-yne (3) Ethyne (4) tert-butyl alcohol

PO0082

46. Which one of the following tests used for the identification of functional groups in organic compounds does not use copper reagent ? **[JEE MAIN 2021]**

(1) Barfoed's test

(2) Seliwanoff's test

(3) Benedict's test

(4) Biuret test for peptide bond

PO0083

INTEGER TYPE

47. Using the provided information in the following paper chromatogram : **[JEE MAIN 2020]**

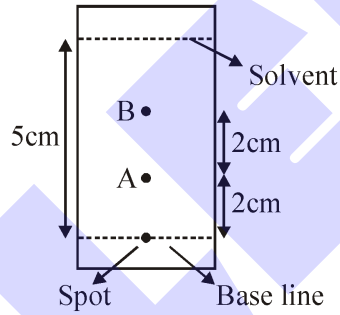


Figure : Paper chromatography for compounds A and B.

the calculate R_f value of A _____ $\times 10^{-1}$. **PO0066**

48. 0.8 g of an organic compound was analysed by Kjeldahl's method for the estimation of nitrogen. If the percentage of nitrogen in the compound was found to be 42%, then _____ mL of 1 M H_2SO_4 would have been neutralized by the ammonia evolved during the analysis. **[JEE MAIN 2021]**

PO0067

49. When 0.15 g of an organic compound was analyzed using Carius method for estimation of bromine, 0.2397 g of AgBr was obtained. The percentage of bromine in the organic compound is _____. (Nearest integer) **[JEE MAIN 2021]**

[Atomic mass : Silver = 108, Bromine = 80] **PO0068**

50. Which one of the following techniques is not used to spot components of a mixture separated on thin layer chromatographic plate? **[JEE MAIN 2022]**

(A) I_2 (Solid)

(B) U.V. Light

(C) Visualisation agent as a component of mobile phase

(D) Spraying of an appropriate reagent **PO0101**

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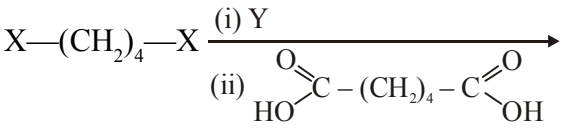
EXERCISE # J-ADVANCED_(OBJECTIVE)

1. Among cellulose, poly(vinyl chloride), nylon and natural rubber, the polymer in which the intermolecular force of attraction is weakest is **[JEE 2009]**

(A) Nylon
(B) Poly (vinyl chloride)
(C) Cellulose
(D) Natural Rubber

BM0161

2. The correct functional group X and the reagent/reaction conditions Y in the following scheme are **[JEE 2011]**



(A) X = $COOCH_3$, Y = H_2 / Ni / heat
(B) X = $CONH_2$, Y = H_2 / Ni / heat
(C) X = $CONH_2$, Y = Br_2 / NaOH
(D) X = CN, Y = H_2 / Ni / heat

BM0165

3. The total number of lone-pairs of electrons in melamine is : **[JEE 2013]**

BM0182

4. Choose the correct option(s) from the following **[JEE 2019]**

(A) Natural rubber is polyisoprene containing *trans* alkene units
(B) Nylon-6 has amide linkages
(C) Cellulose has only α-D-glucose units that are joined by glycosidic linkages
(D) Teflon prepared by heating tetrafluoroethene in presence of a persulphate catalyst at high pressure

BM0172

5. Match the compounds in LIST-I with the observation in LIST-II, and choose the correct option.

LIST-I	LIST-II
(I) Aniline	(P) Sodium fusion extract of the compound on boiling with $FeSO_4$, followed by acidification with conc. H_2SO_4, gives Prussian blue color.
(II) o-Cresol	(Q) Sodium fusion extract of the compound on treatment with sodium nitroprusside gives blood red color.
(III) Cysteine	(R) Addition of the compound to a saturated solution of $NaHCO_3$ results in effervescence.
(IV) Coprolactam	(S) The compound reacts with bromine water to give a white precipitate.
	(T) Treating the compound with neutral $FeCl_3$ solution produces violet color.

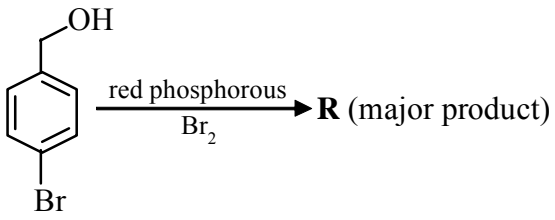
[JEE 2022]

(A) I→P, Q; II→S; III →Q, R; IV→P
(B) I→P ; II→R, S; III→R; IV→Q, S
(C) I→Q, S; II→P, T; III→P; IV→S
(D) I→P, S; II→T; III→Q, R; IV→P

PO0102

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6. Consider the following reaction. **[JEE 2022]**



On estimation of bromine in 1.00 g of R using Carius method, the amount of AgBr formed (in g) is _____.

[Given : Atomic mass of H = 1, C = 12, O = 16, P = 31, Br = 80, Ag = 108]

PO0103

7. Among the following, the correct statement(s) about polymers is(are) **[JEE 2022]**

(A) The polymerization of chloroprene gives natural rubber.

(B) Teflon is prepared from tetrafluoroethene by heating it with persulphate catalyst at high pressures.

(C) PVC are thermoplastic polymers.

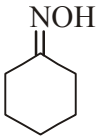
(D) Ethene at 350-570 K temperature and 1000-2000 atm pressure in the presence of a peroxide initiator yields high density polythene. **PO0104**

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EXERCISE # J-ADVANCED_(SUBJECTIVE)

1. Give the structures of the products in each of the following reactions. **[JEE 2000]**

(i) Sucrose $\xrightarrow{H^+}$ A + B

(ii)  $\xrightarrow{H^+}$ C $\xrightarrow{\text{Polymerisation}}$ $[-D-]_n$

BM0173

2. Monomer A of a polymer on ozonolysis yields two moles of HCHO and one mole of CH_3COCHO.

(a) Deduce the structure of A. **[JEE 2005]**

(b) Write the structure of "all cis" – form of polymer of compound A.

BM0177

3. Match the chemical substances in Column I with type of polymers / type of bonds in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS. **[JEE 2007]**

Column I	**Column II**
(A) Cellulose	(P) Natural polymer
(B) Nylon-6, 6	(Q) Synthetic polymer
(C) Protein	(R) amide linkage
(D) Sucrose	(S) Glycoside linkage

BM0156

4. Match the compounds/ion in column I with their properties/ reaction in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS. **[JEE 2007]**

Column I	**Column II**
(A) C_6H_5CHO	(P) gives precipitate with 2,4-dinitrophenylhydrazine
(B) $CH_3C{\equiv}CH$	(Q) gives precipitate with $AgNO_3$
(C) CN^-	(R) is a nucleophile
(D) I^-	(S) is involved in cyanohydrin formation

BM0157

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ANSWER KEY

EXERCISE # O-I

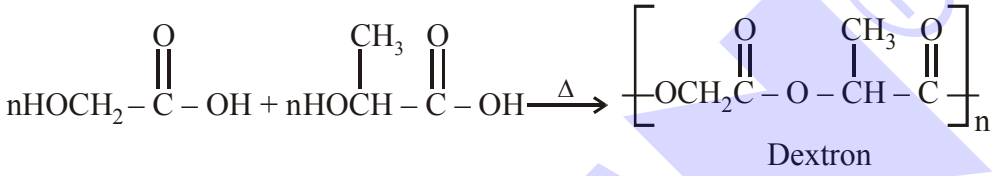
1. (D)	**2. (A)**	**3. (C)**	**4. (C)**	**5. (C)**	**6. (A)**
7. (B)	**8. (A)**	**9. (C)**	**10. (A)**	**11. (B)**	**12. (D)**
13. (D)	**14. (D)**	**15. (D)**	**16. (B)**	**17. (A)**	**18. (B)**
19. (A,B,C,D)	**20. (A,B,C)**				

21.(D)

Sol. (A) Nylon–2–Nylon–6

(B) PHBV

(C) Dextron :



Dextron

(D) Nylon-6,6

22. (A,C)	**23. (A,B)**	**24. (B)**	**25. (B)**	**26. (D)**

27. (B)

Sol. Neoprene is elastomer
Nylon-6, 6 is fiber
PVC is thermoplastic
Bakelite is thermosetting

28. (C)

Sol. Low density polyetne is used in the insulation of electricity carrying wires and manufacture of squeeze bottles, toys and flexible pipes.

29. (A)	**30. (A,B,C)**	**31. (B,C,D)**	**32. (B)**	**33. (C)**	**34. (B)**
35. (B)	**36. (C)**	**37. (C)**	**38. (B)**	**39. (B)**	**40. (A,B,C,D)**

EXERCISE # O-II

1. (4)	**2. (4)**	**3. (2)**	**4. (4)**	**5. (3)**	**6. (3)**
7. (1)	**8. (2)**				

9. (1)

Sol. Theory based

10. (2)

Sol. Theory Based

11. (4)

12. (2) **13. (2)**

14.(3)

Sol. $Na + C + N \xrightarrow{\Delta} NaCN$

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15. (4) **16. (2)**

17.(2)

Sol. Organic compound containing N and S will give red color on addition of $FeCl_3$.

18. (3) **19. (4)**

20.(4)

Sol. Theory Based

21. (4) **22. (3)** **23. (1)**

24. (2)

Sol. $R_F = \dfrac{\text{distance travelled by the compound}}{\text{distance travelled by the solvent}}$

for X ; $R_f = \dfrac{14.3}{18} = 0.79$

25. (3)

26. (2)

Sol. Organic compound containing only N not S will give blue color ppt.

27. (2) **28. (3)** **29. (2)** **30. (1)** **31. (3)**

32. (3) **33. (4)**

34. (1)

Sol. Equivalent of NH_3 evolved $= \dfrac{100\times0.1\times2}{1000} - \dfrac{20\times0.5}{1000} = \dfrac{1}{100}$

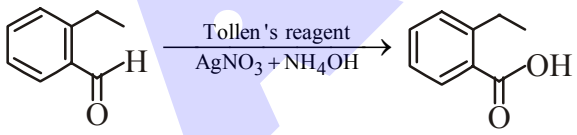
% of N in the unknown O.C. $= \dfrac{1}{100}\times\dfrac{14}{0.3}\times100 = 46.6\%$

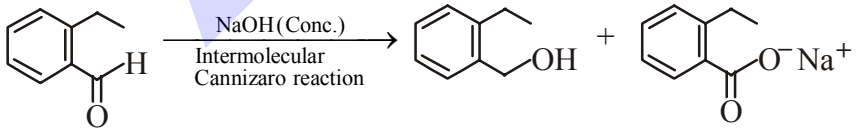
% of N in urea $(NH_2)_2Co = \dfrac{14\times2}{60}\times100 = 46.6\%$

35. (3)

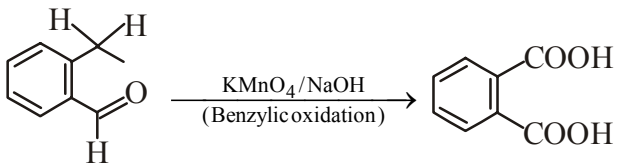
36. (3)

Sol.





No α–H

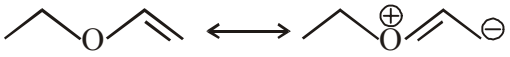


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37. (4)

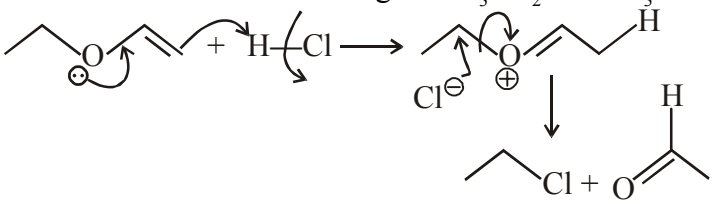
38.(2)

Sol.



Unsaturation present

When react with conc. HCl to give $CH_3CH_2Cl + CH_3CHO$



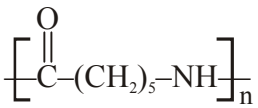
39.(1)

Sol. Compound C is the most nonpolar compound because it contains only hydrogen and carbon. Compounds A and B are more polar because of the presence of oxygen and hence the presence of polarised carbon-oxygen bonds. The alcohol group of compound B makes this compound the most polar of the three molecules by virtue of hydrogen bonding capabilities as well as the carbon-oxygen dipole. Compound B would thus elute last.

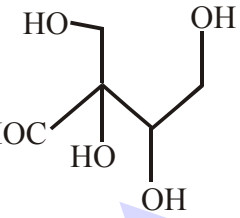
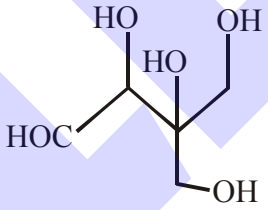
40. Ans.(B,D)

EXERCISE # S-I

1. (C) **2. (A)** **3. (C)** **4. (A)→P, Q, R ; (B) → S ; (C)→R ; (D) → P, Q, S**

5. (a)  **(b)** theory based

6. (A) → Q, R (B) → P, (C) →S; (D) →T **7.(A) → R ; (B) → P ; (C) → S ; (D) → Q**

8.  **or** 

9. (16.67)

Sol. $\underset{0.38\,m}{O.C.} \xrightarrow{\Delta} \underset{50\,mL}{N_2}$

T = 300 K

P = 715 mm Hg

aq. tension = 15 mm Hg

P of N_2 gas = 715 – 15 = 700 mm Hg

$$n_{N_2} = \frac{PV}{R_T}$$

$$n_{N_2} = \frac{\frac{700}{760} \times \frac{50}{1000}}{0.0821 \times 300}$$

$$n_{N_2} = 0.0018$$

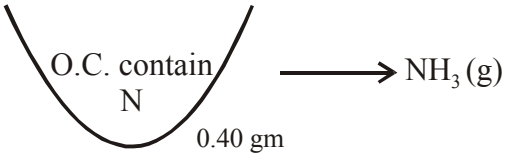
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\S-Carboxylic-Aromatic-Bismole-Poly, POC & Sep. Tec-Che E-very day\Eng\4-Poly & POC\03_Poly-POC-Sepr_Ex.p65

wt. of nitrogen = $0.0018 \times 28 = 0.05$

% of N in O.C = $\frac{0.05}{0.30} \times 100 = 16.67\%$

10. (70.00)

Sol.



$$2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$$

50 ml

20 m mole 0.5 M

25 M mole

–10 M mole

15 M mole

Left $H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$

15 M mole 60 ml, 0.5 M

30 M mol

Moles of NH_3 = 20 M mol = 0.02 mol

Moles of N = 0.02

Wt. of N = $0.02 \times 14 = 0.28$

$$\% N = \frac{0.28}{0.40} \times 100$$

%N = 70

11. (6)

O.C. containing N and S give blood red ppt of $[Fe(SCN)]^{2+}$ when treated with neutral $FeCl_3$ solution.

(i), (ii), (iii), (v) (viii), (ix)

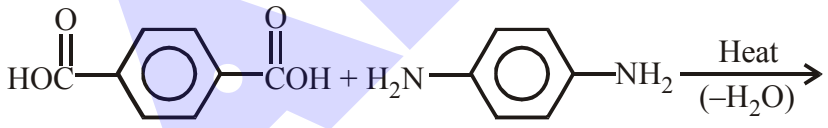
12. (B)

13. (D)

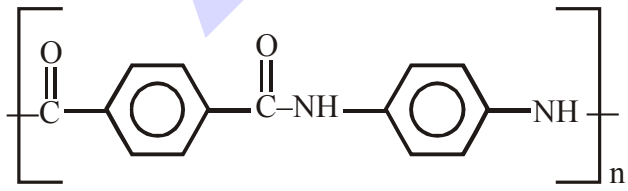
14. (A)

15. (A,B,C,D)

Sol.



Terephthalic acid p-Phenylene diamine



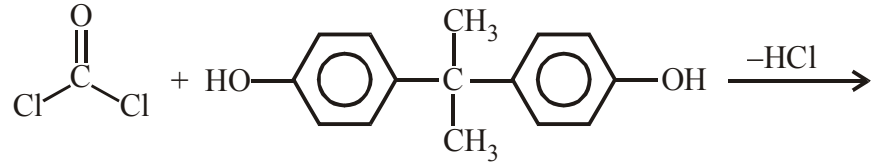
Kevlar

Aromatic polyamides are known as aramids. Kevlar is an aramide. Kevlar fibers are very strong and have have tensile strength greater than that of steel. It is used for making army helmets and ligth weight bullet proof vests.

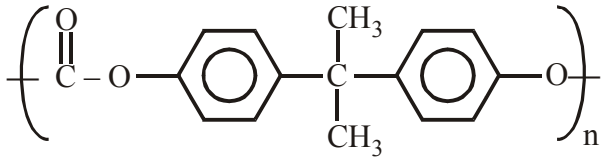
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\S-Carboxylic+Aramatic Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\4-Poly & POC\03_Poly-POC-Sep_Ex.p65

16. A,B,C,D

Sol.



Phosgene Bisphenol A



Lexan (A polycarbonate polymer)

Polycarbonates are like polyester but their carbonyl group is linked to two alkoxy or aryloxy groups.
Lexan is prepared by the reaction of phosgene with bisphenol A in the presence of pyridine.
Lexan is a strong and transparent polymer used for bullet-proof windows and traffic light lenses, compact discs and two-wheeler safety helmets.

17. A → R, B → S, C → T, D → Q

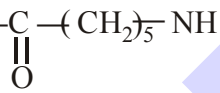
18.(4)

Sol. (i), (ii), (iii), (iv)

19.(5)

Sol. 2, 3, 4, 5, 6

20. 7

Sol. Mass of one  unit is = 113

$(113 \times x) + 18 = 809$ ($x \Rightarrow$ number of monomer units)

$113 \times x = 791$

$x = 7$

EXERCISE # J-MAIN

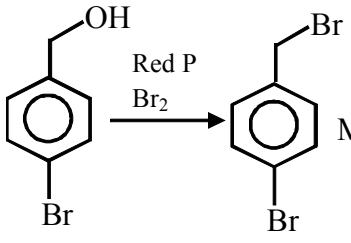
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	2	4	4	1	2	1	2	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	1	2	4	1	1	2	4	3	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	2	1	2	3	1	2	4	3	3
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	2	3	1	2	3	1	2	3	1	2
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	3	4	1	4	2	2	4	12	68	3

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\S-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\4-Poly & POC\03_Poly-POC-Sepr_Exp65

EXERCISE # J-ADVANCED_(OBJECTIVE)

1. (D) **2. (C,D)** **3. (6)** **4. (B,D)** **5. (D)**

6. (1.50)

Sol. 

M.W. = 250 g/mol

(R)

$1g\ R \rightarrow \frac{1}{250} moles$

$No.\ of\ Br\ Atoms \rightarrow \frac{2}{250} moles$

$Moles\ of\ AgBr \rightarrow \frac{2}{250} moles$

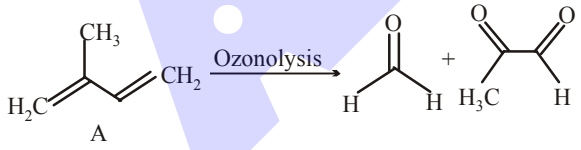
$Mass\ of\ AgBr = \frac{2}{250} \times (108+80) = 1.504$

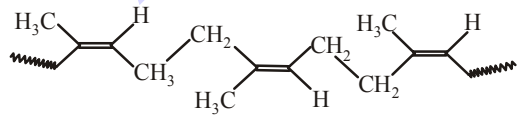
7. (B,C)

Sol. (a) The polymerisation of neoprene gives natural rubber.
(b) is correct statement
(c) is correct statement
(d) Ethene at 350-570 K temperature and 1000-2000 atm pressure in the pressure of a peroxide initiator yields low density polythene.

EXERCISE # J-ADVANCED_(SUBJECTIVE)

1. (i) α-D–glucose and β-D–fructose
(ii) C = Caprolactum, D = Nylon-6

2. (a) 

(b) 

3.(A) → P, S ; (B) → Q, R ; (C) → P, R ; (D) → S

4.(A) → P, S ; (B) → Q ; (C) → Q, R, S ; (D) → Q, R

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